AI Agent Platform for Company STU-MSFT

You are an AI engineering team supporting **STU-MSFT** company in designing and building an **AI Agent Platform as a Service (PaaS)**.

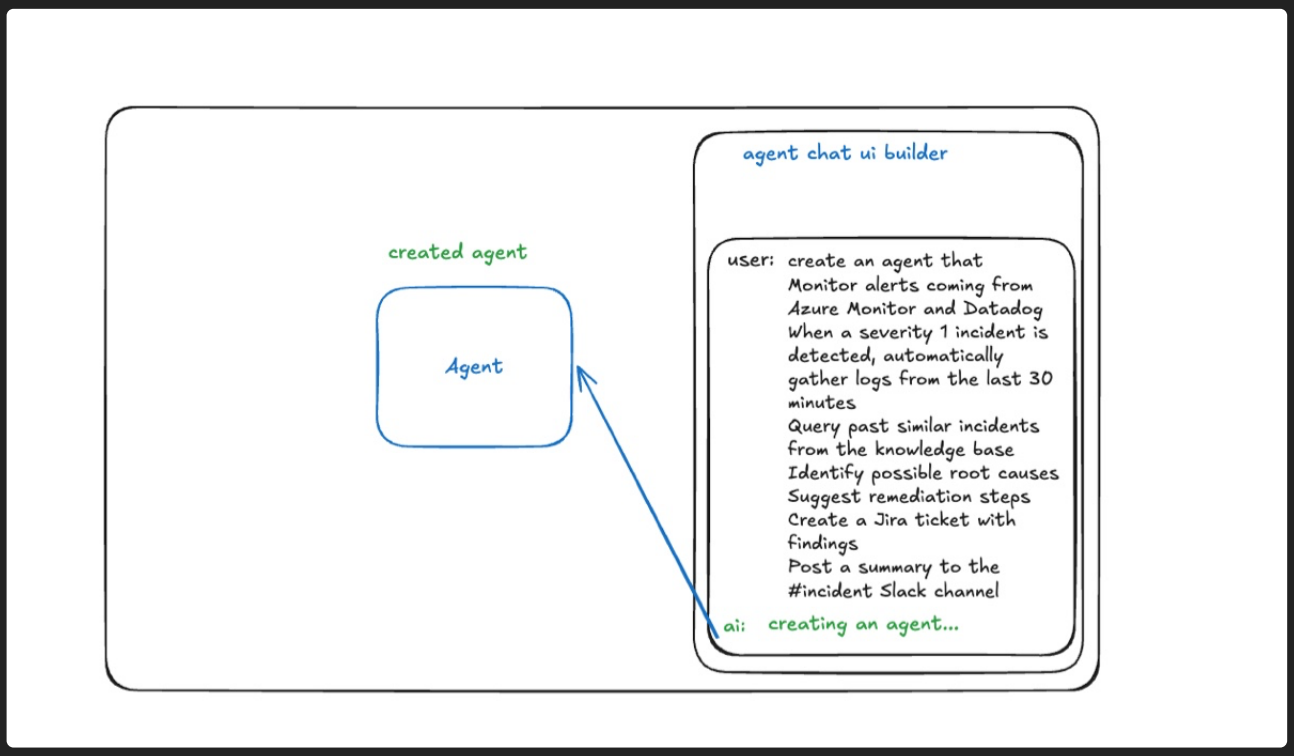
The platform should enable product teams to:

- Create and configure AI agents through a user-friendly UI
- Attach and manage tools for each agent
- Connect and manage multiple data sources
- Support parallel execution and sub-agent orchestration
- Ensure secure, isolated runtime environments for every agent
- Connect agents together to build complete workflows, including both sequential and autonomous execution flows
- Terminal & CLI Execution
- Evaluation & Quality - for each of the agents
- Cost & Token Observability
- Model Abstraction & Routing Layer
- Agent marketplace
- Memory management - Long & Short term memory

What you need to design:

- Agent control plane
- Agent runtime plane
- Model abstraction layer
- Multi model routing
- Memory management - long & short term
- Thread management
- State Management
- Tool marketplace
- Policy engine
- Evaluation Engine
- Cost observability dashboard
- The system must be designed to operate at large scale

Demo of the UI:



What you need to prepare to the session

1. **HLD - High-Level System Design**

Provide a comprehensive high-level architecture design (without referencing any specific products or vendors).

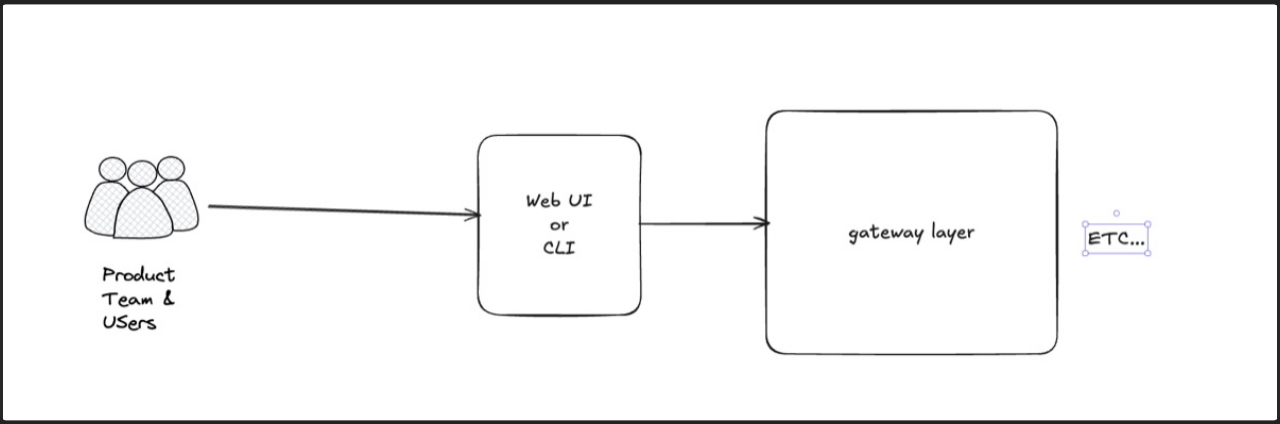
The design must cover all technical aspects of the platform, including

- A clearly defined **Control Plane**
- A fully detailed **Runtime (Data) Plane**

The HLD should go deep into the technical architecture, covering system components, interactions, data flows, security boundaries, memory usage, scalability considerations, isolation mechanisms, observability, fault tolerance, and governance controls, rag systems.

The expectation is a thorough architectural blueprint that reflects production-grade, large-scale system design thinking.

HLD example (a minimal example) :



2. **Product-Based Architecture Design**

   Based on the HLD, design a complete product-specific architecture that maps the logical components into concrete Microsoft technologies (leveraging Microsoft products as extensively as possible).

   The design should:

   - Translate the Control Plane and Runtime Plane into specific Microsoft services
   - Define how each component is implemented using Microsoft technologies (Use Microsoft products as extensively as possible for this use case)
   - Include detailed system flows (end-to-end request lifecycle, agent execution flow, memory management, thread management, policy enforcement, evaluation, etc.)
   - Describe data movement, integrations, security enforcement, scalability model, and observability
   - Reflect a production-ready, enterprise-grade deployment architecture
   - Ensure observability, evaluation workflows, and cost management are taken into account.

3. **Build a POC**

   Using the architecture you designed in the previous stages, build a working Proof of Concept (PoC) that demonstrates the platform's capabilities.

The PoC should allow users to interact through either a CLI or a UI to create and configure an example agent, showcasing the end-to-end flow from agent definition to execution.